

OrbitWatch

Space traffic control: a live globe of satellites and debris, with voice briefings on close approaches.

The Idea

A 3D globe showing thousands of real satellites and pieces of debris moving in real time. OrbitWatch screens for objects that will pass close to each other and Grok Voice delivers a “flight director” briefing: *“Three close approaches in the next 24 hours. The closest is...”* You can also ask by voice: *“Show me every Starlink satellite over Europe.”*

Track Fit

Built with Cursor	Globe rendering, orbit propagation, and screening logic built in Cursor.
Grok Voice / Imagine	Voice for briefings and voice queries.
Real space data	CelesTrak orbital data for active satellites and debris.
Navigation theme	Navigating increasingly crowded orbits over the next 100 years.
Also eligible	Big Red, Software, Design.

Features

MVP

- 3D globe with live positions of active satellites (a few thousand objects).
- Filter by group (Starlink, ISS, weather, debris).
- Close-approach screening for one chosen satellite against all others over the next 24 hours.
- Spoken Grok briefing summarizing the results.

Stretch

- Voice queries that change the globe view.
- Suggest a simple avoidance maneuver (raise or lower the orbit).
- Timeline scrubber to replay the past week.
- Add the full debris catalog.

Tech Stack and Data

- Frontend: CesiumJS or three.js / globe.gl.
- Orbits: CelesTrak GP data, propagated with `satellite.js` (SGP4).
- Screening: Python or a Web Worker; coarse filter, then fine time-step check.
- AI: Grok Voice API.

Limitations and Risks

- **Not real collision risk.** Public orbital data (TLEs) is only accurate to around a kilometer or more and degrades over days. Real conjunction assessment uses error estimates (covariance) that are not public.
Mitigation: Call the output “close-approach screening”, not collision prediction. State this in the pitch; judges will respect it.
- **Scale.** Tens of thousands of tracked objects means checking every pair is billions of comparisons.
Mitigation: Screen only one chosen satellite against all others, and use a cheap altitude-band filter first.
- **Rendering performance.** Animating thousands of points on a 3D globe can drop frames on laptops.
Mitigation: Use instanced points, update positions a few times a second, and cap the default object count.
- **Originality.** Existing visualizers (stuffin.space, CelesTrak SOCRATES, LeoLabs) already do much of this.
Mitigation: Differentiate with the voice “flight director” and natural-language queries.
- **Weak use of Grok Imagine.** Only Voice fits naturally; Imagine would feel forced.
Mitigation: The track requires Imagine *or* Voice, so Voice alone is enough.
- **CelesTrak usage limits.** CelesTrak asks users not to download data too often.
Mitigation: Download once, cache locally, refresh at most every few hours.

Scorecard

Demo-able	4/5	The live globe is eye-catching.
Buildable in 24h	3/5	Globe is easy; screening takes care.
Navigation theme	4/5	Good, slightly indirect.
Wow factor	4/5	Thousands of moving satellites.
Technical risk (5 = riskiest)	3/5	Moderate (performance, scale).

Verdict

Solid option. Visually impressive and data-heavy, which suits the SpaceX track. The main weakness is that similar tools already exist, so the voice experience must carry the originality.